

NATIONAL INSTITUTE OF TECHNOLOGY DURGAPUR
Odd Semester Mid-term Examination, 2025-2026

Course Code: CEC303

Course Name: Fluid Mechanic

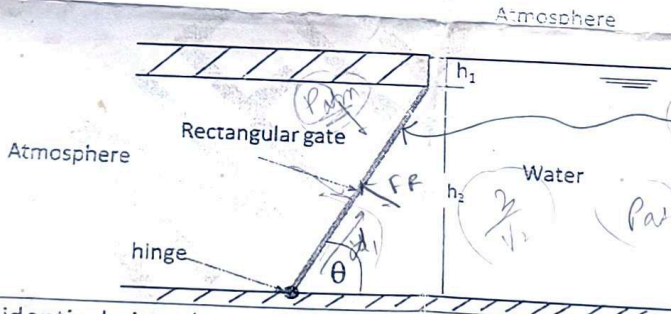
Full Marks: 25

Time: 90 Mins

Question Paper No.: NITDGP/CEC303/01

Date of Exam: 22/09/2025

Instructions: Answer all the questions.

Question No.	Body of the Question	Marks	Mapped CO
1	a) Differentiate between pathline, streakline, and streamline. Under what condition do they coincide? b) Explain the concept of fully developed flow in a circular pipe. Why is it important in design? c) Define the Hydraulic Grade Line (HGL) and Energy Grade Line (EGL). How do frictional losses in a pipe affect the slope of the HGL and EGL? Explain how minor losses (valves, bends, fittings) appear in the HGL and EGL. d) Draw the velocity profile for laminar and turbulent flow in a circular pipe. e) Explain the difference between gauge pressure, absolute pressure, and vacuum pressure. f) Define Metacenter and Metacentric height. g) What is the Moody diagram? Explain its uses. Name and describe the main regions or parts of the Moody diagram. h) What is the no-slip boundary condition? Explain its significance in fluid flow over solid surfaces.	2 2 2 2 2 1 3 1	CO CO2 CO1 CO3 CO1 CO2 CO1 CO3
	In the given figure below, a rectangular gate is hinged at the bottom and inclined into a wall at an angle of 45°. Consider the width of the gate as 1 m, with $h_1=1$ m and $h_2=3$ m. Calculate whether the gate will stay in place or fall.	4	CO3
			
	Two identical pipes (i.e. having the same length, same diameter, and same roughness) are used to withdraw water from a reservoir. In the first case, they are attached in series and also discharge freely into the atmosphere. In the second case, they are attached in parallel. Ignore the minor losses and consider the friction losses is same in both cases. Find out the ratio of the discharge in the parallel arrangement to that in the series arrangement.	4	CO2
	Find the kinematic viscosity of an oil having density of 981 kg/m^3. The shear stress at a point in oil is 0.2452 N/m^2 and velocity gradient at that point is 0.2 per second	2	CO1

Course Outcomes

- CO1: Development of skills for predicting fluid behaviour
- CO2: Knowledge of basics of fluid flow measurement and model development
- CO3: Developing the requisite skill that helps in studying the advanced courses

